Kevin Zhang

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SUMMARY

Research ML Scientist with a **Ph.D. in Statistics** working at the intersection of large language models and quantitative modeling. Current research spans **AI safety** (jailbreak detection, adversarial robustness, agentic AI) and **conformal uncertainty quantification** for generative models (ICML 2026), alongside ML-driven optimization of a $300B fixed-income portfolio. Grounded in stochastic differential equations, optimal transport, and generative modeling, with production ML experience at fintech scale.

EDUCATION

Doctor of Philosophy (PhD) *statistics* — 2019 - 2024
University of Toronto, Toronto, Canada
Supervisors: Dr. Dehan Kong and Dr. Zhaolei Zhang
Dissertation focused on generative modeling, stochastic differential equations, and optimal transport methods.

Honours Bachelor of Science (HBSc) *pathobiology, statistics, mathematics* — 2015 - 2019
University of Toronto, Toronto, Canada — major GPA: 4.0/4.0, GPA: 3.9/4.0

WORK EXPERIENCE

Research Machine Learning Scientist II, Layer 6 AI (TD Bank Group) — July 2025 - Present

- **AI Safety**: Lead applied and theoretical LLM safety research with expertise in jailbreak detection, adversarial robustness, and agentic AI, and translate it into safeguards for enterprise LLM systems.
- **Uncertainty Quantification**: Developed conformal prediction methods for generative models, providing distribution-free, finite-sample coverage guarantees on generative and LLM outputs (ICML 2026).
- **Quantitative Modeling**: Built an ML use case for bond portfolio optimization on a $300B fixed-income portfolio.
- **Delivery**: Ship internal and public-facing AI tools at continental scale; contribute to patents and publications.

Software Developer, Robinhood — October 2024 - July 2025

- **Fraud & Security**: Built authentication pipelines and ML-based fraud detection, cutting fraud losses by millions of dollars.
- **ML at Scale**: Partnered with data science to deploy large-scale image and video models across crypto, options, and securities markets; integrated LLM-powered tooling into the customer support stack.
- **Systems**: Built and maintained low-latency, high-availability API services using Go, Python, Django, gRPC, Kubernetes, Docker, Bazel, and Jenkins, ensuring reliable CI/CD.

Sessional Lecturer, University of Toronto — May 2024 - June 2024

- Instructed STA237 (Probability, Statistics, and Data Analysis I) to 200+ students; led a team of teaching assistants.

SKILLS

- **Languages**: Python, Go, C++, SQL, R
- **ML & LLMs**: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, LangGraph, CUDA
- **Quantitative Methods**: Stochastic differential equations, optimal transport, conformal prediction, time series & anomaly detection, portfolio optimization
- **Infrastructure**: Docker, Kubernetes, Bazel, AWS, Databricks, SLURM, Git, Linux

SELECTED PUBLICATIONS

- Loaiza-Ganem, G., Zhang, K., Cui, W., Law, M.T. and Leung, K.K., 2026. "Conf-Gen: Conformal Uncertainty Quantification for Generative Models". *International Conference on Machine Learning (ICML)*.
- Zhu, J., Zhang, K., Kong, D. and Zhang, Z., 2026. "LLOT: Application of Laplacian Linear Optimal Transport in Spatial Transcriptome Reconstruction". *Biometrics*, 82(1).
- Zhu, J., Zhang, K., Zhang, Z. and Kong, D., 2025. "Modeling Cell Type Developmental Trajectory using Multinomial Unbalanced Optimal Transport". *arXiv preprint* arXiv:2501.03501. [Link to article][Link to GitHub]
- Zhang, K., Zhu, J., Kong, D. and Zhang, Z., 2024. "Modeling Single Cell Trajectory Using Forward-Backward Stochastic Differential Equations". *PLOS Computational Biology*. [Link to article][Link to GitHub]